

GUIDED LESSON · LEARN BY EXPERIMENTING

Phase & Interference

A from-scratch walkthrough for total beginners: what a qubit is, why probability alone can lie to you, and how the Z gate makes phase visible — with worked numbers and circuit diagrams.

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Before You Begin

Four ideas you need first

1. A qubit is not a coin

A classical bit is 0 **or** 1. A **qubit** can be in a **superposition** — a bit of both $|0\rangle$ and $|1\rangle$ at once — described by two complex numbers called **amplitudes**, one attached to each basis state.

2. Amplitude → probability

You never observe an amplitude directly. To get the probability of measuring a basis state, you **square its amplitude's magnitude**. Example: an amplitude of 0.7071 gives a probability of $0.7071^2 \approx 0.5$, i.e. a 50% chance.

3. A gate is just a matrix

A **quantum gate** is a small table of numbers (a matrix) that multiplies the qubit's two amplitudes to produce two new amplitudes. Gates must be reversible — you can always undo them — unlike many classical logic gates.

4. How to read a circuit diagram

A circuit is drawn as a horizontal line (the qubit, moving through time left to right) with boxes on it. Each box is a gate; you apply them in the order they appear. The label before the wire is the starting state, the label after is the result.

q0

 $|0\rangle$

H

 $\rightarrow (|0\rangle + |1\rangle)/\sqrt{2}$

Example: a Hadamard gate turns $|0\rangle$ into an equal superposition.

01

Look Beyond Probabilities

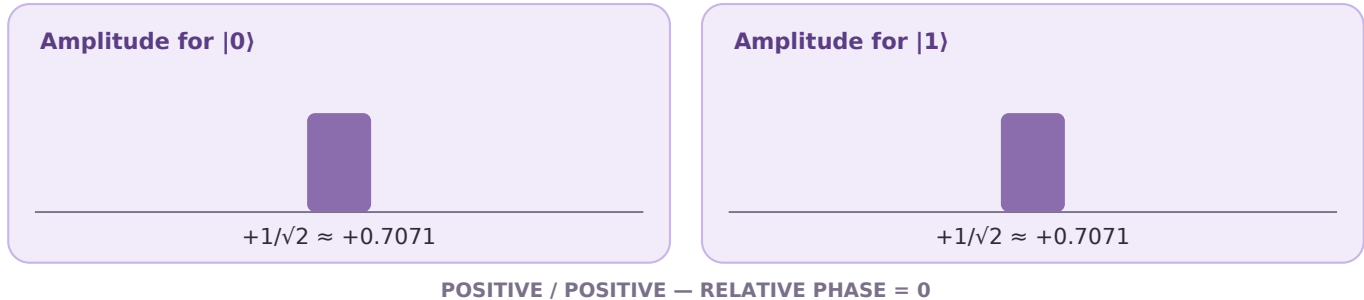
Amplitude, sign, and relative phase

A qubit has two basis states, $|0\rangle$ and $|1\rangle$. An **amplitude** is the number attached to each basis component. Squaring its magnitude gives that component's measurement probability. A **statevector** is simply the ordered list of these amplitudes.

Why probability alone isn't enough

Positive and negative amplitudes can carry the exact same probability: $(+0.7071)^2$ and $(-0.7071)^2$ are both ≈ 0.5 . Two genuinely different states can therefore produce identical probability bars — probability alone doesn't tell the full story.

CONCEPT SKETCH · NOT A SIMULATOR RESULT



WORKED EXAMPLE — TWO STATES, ONE PROBABILITY TABLE

State A = $(|0\rangle + |1\rangle)/\sqrt{2} \rightarrow$ amplitudes $(+0.7071, +0.7071) \rightarrow$ probabilities $(0.5, 0.5)$

State B = $(|0\rangle - |1\rangle)/\sqrt{2} \rightarrow$ amplitudes $(+0.7071, -0.7071) \rightarrow$ probabilities $(0.5, 0.5)$

Measured directly, A and B are indistinguishable — both give heads-or-tails 50/50. But they are **not the same state**: only B has a minus sign, a **relative phase of π radians**. Section 02 shows a gate that reveals the difference.

GLOBAL vs. RELATIVE PHASE

Multiplying an *entire* state by an overall factor $e^{i\alpha}$ is a **global** phase — it never changes any measurement outcome. Only the **relative** phase *between* amplitudes is physically observable, and it's what drives interference.

TAKEAWAY

Same probabilities $\neq$ same state. Two statevectors with identical probability bars can still differ — and behave differently under further gates — because of relative phase.

02

Meet the Z Gate

The single-qubit phase-flip

The **Z gate** (Pauli-Z) is a single-qubit gate that leaves probabilities untouched but flips relative phase — it acts on exactly the property Section 01 introduced.

PAULI-Z MATRIX

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$|0\rangle \rightarrow |0\rangle$$

$$|1\rangle \rightarrow -|1\rangle$$

What it does

- **|0⟩ is unchanged.** The gate leaves the $|0\rangle$ component exactly as it is.
- **|1⟩ picks up a minus sign** — becoming $-|1\rangle$ — a phase kick of π radians on that component only.
- Because $|-1|^2 = 1$, **no probability changes** on its own: measuring right after a Z gate looks identical to measuring before it.
- On the Bloch sphere, Z is a **180° rotation about the z-axis** — the vector spins in place around the pole line joining $|0\rangle$ and $|1\rangle$.

WORKED EXAMPLE — APPLYING Z BY HAND

Start from State A above: $|\psi\rangle = (+0.7071)|0\rangle + (+0.7071)|1\rangle$.

Z multiplies the $|0\rangle$ amplitude by **+1** and the $|1\rangle$ amplitude by **-1**:

$$Z|\psi\rangle = (+1 \times 0.7071)|0\rangle + (-1 \times 0.7071)|1\rangle = (+0.7071)|0\rangle + (-0.7071)|1\rangle = \text{State B.}$$

One gate, and State A becomes State B — same probabilities throughout, phase flipped.

q0

$$(|0\rangle + |1\rangle)/\sqrt{2} \xrightarrow{\text{Z}} (|0\rangle - |1\rangle)/\sqrt{2}$$

Still 50% / 50% if measured now — the change is invisible until you interfere it.

Making the phase flip visible: H → Z → H

On its own, Z's effect can't be seen in measurement probabilities. Sandwich it between two Hadamard gates and the invisible phase flip turns into a visible probability flip:

q0

$$|0\rangle \xrightarrow{\text{H}} \xrightarrow{\text{Z}} \xrightarrow{\text{H}} |0\rangle \text{ (100\% } \rightarrow \text{ 0)}$$

Without Z: the two Hadamards cancel — you always measure 0.

q0



With Z in between: same start, opposite certain outcome.

WHY THIS HAPPENS (SHORT VERSION)

H spreads $|0\rangle$ into the equal superposition $(|0\rangle + |1\rangle)/\sqrt{2}$. Z flips that to $(|0\rangle - |1\rangle)/\sqrt{2}$. The second H then recombines the amplitudes — and because one sign flipped, the amplitudes that used to add up now cancel for $|0\rangle$ and reinforce for $|1\rangle$. That cancel/reinforce pattern is **interference**, and it's only possible because Z changed the relative phase.

Where else it shows up

- Sits alongside the X gate as one of the basic single-qubit tools used to steer a state between $|\psi^+\rangle$, $|\psi^-\rangle$, $|\phi^+\rangle$ and $|\phi^-\rangle$ in multi-qubit circuits.
- Applying Z to one qubit of an entangled pair — combined with X where needed — is a standard way to move between the four Bell states.

TAKEAWAY

Z is phase made visible-in-waiting: silent on its own, decisive once interference gets a chance to turn a phase difference into a probability difference.